AKASH KUNDU

AI/ML Engineer · Computer Vision · NLP · Predictive Modeling
akashkundu1152@gmail.com | +91-9064349004 | akashkundu.me | LinkedIn | GitHub

SUMMARY

I build AI systems that ship. Cut model training time by 70%, achieved sub-50ms inference on a locally-deployed LLM assistant, and classified 7 eye diseases at >85% accuracy - all as a 3rd-year undergrad. I work across the full ML pipeline: data preprocessing and feature engineering through model evaluation and containerised deployment on Azure with Docker. Proven team lead under pressure: served as team lead for Smart India Hackathon in both 2024 and 2025, coordinating a 6-member cross-functional team through end-to-end problem scoping and prototyping. Graduating September 2027.

EDUCATION

B.Tech — Computer Science & Engineering 08/2023 – 09/2027 (Expected)
Techno India University, Kolkata *CGPA: 7.5 / 10 (through 5th Semester)*

Higher Secondary — Science (Class XII) 04/2021 – 05/2023
Arambagh Vivekananda Academy *Percentage: 71.6%*

EXPERIENCE

AI & Machine Learning Intern (Supported by Google for Developers) 10/2025 – 12/2025
AICTE & EduSkills *Virtual*

- Completed 100% of assessed milestones across a 10-week Google-supported ML curriculum - covering NLP, predictive modeling, and neural networks - by building hands-on Python experiments in scikit-learn and TensorFlow with structured documentation of findings per training run.
- Developed and evaluated AI-powered automation workflows across 4+ module assessments - covering feature engineering, model accuracy optimisation, and dataset experimentation - applying the full ML pipeline from raw data preprocessing through to performance reporting.

PROJECTS

OphthalmoAI: AI-based Eye Disease Predictor 01/2026 – 04/2026
Python · PyTorch · OpenCV · Pandas · Grad-CAM *Completed*

- >85% classification accuracy across 7 ocular diseases - measured by held-out test-set evaluation - by fine-tuning an EfficientNet CNN in PyTorch with structured feature engineering and data augmentation applied to 5,000+ images via a Pandas preprocessing pipeline.
- 70% reduction in epoch training time - benchmarked against CPU baseline - by migrating the full training workflow to GPU-accelerated environments, enabling faster experiment iteration and model evaluation cycles.
- 40% reduction in manual clinical review time - measured per-report - by automating PDF diagnostic output with Grad-CAM visualisations that highlight disease regions, producing interpretable and documented model findings.

COPPER: AI Desktop Productivity Assistant 03/2026 – Present
Tauri · FastAPI · Ollama · Local LLMs *Actively Developing*

- 30% improvement in daily workflow efficiency - measured against manual task baselines - by building an AI-powered automation tool that runs NLP inference 100% offline via Tauri and FastAPI with sub-50ms latency, with zero third-party API dependency.

- 10+ daily voice queries processed with zero data leakage - verified by end-to-end offline execution logs - by implementing an asynchronous voice-to-text NLP pipeline backed by a persistent local memory layer.

SpecBridge: AI Hardware Benchmark Engine 02/2026 – Present

Python · FastAPI · CatBoost · Docker *Actively Developing*

- 95% prediction reliability across 3 hardware workload types - built with CatBoost on an automated feature engineering pipeline that scraped and structured data from 6+ hardware specification tables.
- Sub-30ms recommendation response time under production load - achieved by containerising the FastAPI serving layer with Docker, ensuring consistent environment deployment and reproducible model evaluation across hardware configurations.

StealthChat: E2EE Multi-User Chat Application 11/2025 – 01/2026

React · Node.js · Java · MongoDB *Completed*

- 50+ concurrent users supported with sub-100ms message delivery - verified under load testing - by engineering a multi-threaded Java and Node.js backend with WebSocket layers, 256-bit E2E encryption, and JWT authentication; demonstrates systems thinking applied to real-time AI serving architectures.
- 10+ active chat rooms with zero message loss - measured across continuous uptime sessions - by integrating MongoDB for persistence and Cloudflare/Ngrok tunnelling for reliable cross-network delivery.

LEADERSHIP

Team Lead · Smart India Hackathon 2024 & 2025 College Round · Both Years

6-member cross-functional team · Techno India University

- Led a 6-member cross-functional team through college-level SIH participation in consecutive years - owning end-to-end problem scoping, AI solution design, and live presentation under sprint conditions; coordinated both technical architecture decisions and team workflow across both editions.

SKILLS

Languages: Python, Java, JavaScript (ES6+), C++, SQL
AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, CatBoost, NLP, Computer Vision (CNNs, EfficientNet), Local LLMs (Ollama), Predictive Modeling, Feature Engineering, Pandas
Model Evaluation: Accuracy assessment, Hyperparameter tuning, Grad-CAM, Data visualisation
Web & Model Serving: FastAPI, Node.js, React.js, Tailwind CSS, REST APIs, WebSockets
Cloud & Infrastructure: Microsoft Azure, Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, MongoDB
Soft Skills: Team leadership, cross-functional collaboration, technical communication, experiment documentation

CERTIFICATIONS

- Agentic AI Certified Foundations Associate – Oracle (06/2026)
- AI-ML Virtual Internship - AICTE & EduSkills (12/2025)
- Getting Started with Enterprise-grade AI - IBM (08/2024)